

WHD SERIES (300 TO 2,000 CFM)

Heatless Twin-Tower Desiccant Air Dryers

Pressure Swing Adsorption (PSA) Technology Delivering Guaranteed -40°C to -70°C Pressure Dew Point for Ultra-Critical Industrial Applications



-40°C PDP

ISO 8573-1 Class 2
Moisture Purity

PSA CYCLE

Automated 8-Minute
Cycle Control

ACTIVATED ALUMINA

High Crush-Strength
Spherical Media

<75 DBA

Dual Acoustic
Depressurization
Mufflers

THERMODYNAMIC ARCHITECTURE & ENGINEERING

Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Win Equipments WHD Series twin-tower desiccant air dryers eliminate atmospheric moisture vapor to sub-zero dew points. Designed for critical manufacturing where any trace of moisture causes irreversible process failures—such as pharmaceutical formulation, semiconductor fabrication, and sub-zero pneumatic instrumentation—WHD dryers utilize Pressure Swing Adsorption (PSA) to guarantee -40°C pressure dew point standard (with -70°C optional using 13X molecular sieve).

Working Principle: Compressed air enters through dual angle-seat non-lubricated switching valves and flows upward through the online adsorbing desiccant bed. Microscopic pores in the spherical activated alumina adsorb moisture molecules via physical attraction, reducing water content to less than 0.11 g/m³. Simultaneously, 12–15% of dried effluent air is depressurized to atmospheric pressure and routed downward through the offline tower, regenerating the desiccant bed and venting vapor via heavy-duty exhaust silencers. Every 4 minutes, the solid-state controller rotates the adsorption and regeneration roles.

SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Air Compressor → Wet Receiver → Pre-Filter Grade X (1μ) → Coalescing Filter Grade Y (0.01μ) → WHD Desiccant Dryer (-40°C PDP) → Particulate Dust Filter Grade P (1μ) → Dry Buffer Vessel → Ultra-Clean Process Header

High Crush-Strength Spherical Activated Alumina

Premium 4–8mm spherical desiccant with exceptional surface area (>340 m²/g) and high mechanical attrition resistance. Prevents desiccant dusting and bed compaction even under violent pneumatic load pulsations.

High-Cycle Angle-Seat Pneumatic Valves

Piston-actuated stainless steel angle seat switching valves tested to over 1,000,000 continuous switching cycles. Zero diaphragm failure risk ensures dependable 24/7 tower changeovers.

Dew Point Dependent Switching (DDS) System

Optional ceramic hygrometer monitors real-time dew point. During low inlet moisture loads, tower regeneration is automatically extended, reducing purge air consumption from 15% down to under 5%.

Heavy-Duty Dual Expansion Silencers

Multi-chamber acoustic exhaust silencers with internal sound baffles dampen rapid tower depressurization noise to under 75 dBA, creating a safe, quiet shop-floor environment.

CERTIFIED ENGINEERING RATINGS

Technical Specifications Matrix

Full standard model range & operational parameters

Standard Design Conditions: Operating Pressure: 7.0 to 16.0 bar g | Inlet Air Temp: 35°C to 45°C | Pressure Dew Point: -40°C Standard (-70°C Optional) | Cycle Time: 8 min (4 min Adsorption / 4 min Regeneration) | Vessel Code: IS 2825 / ASME Sec VIII Div 1

MODEL CODE	NOMINAL CFM	FLOW M ³ /HR	RATED PRESSURE	FLANGE CONN	CONTROL SUPPLY	DIMENSIONS (L×W×H MM)	WEIGHT KG
WHD - 030	300 CFM	510	7–16 bar g	1.5" NB Flange	220V / 1Ph / 50Hz	1450 × 1100 × 2500	450 kg
WHD - 040	400 CFM	680	7–16 bar g	2" NB Flange	220V / 1Ph / 50Hz	1450 × 1200 × 2500	630 kg
WHD - 050	500 CFM	850	7–16 bar g	3" NB Flange	220V / 1Ph / 50Hz	1650 × 1250 × 2300	720 kg
WHD - 060	600 CFM	1,020	7–16 bar g	3" NB Flange	220V / 1Ph / 50Hz	1650 × 1250 × 2500	910 kg
WHD - 075	750 CFM	1,275	7–16 bar g	3" NB Flange	220V / 1Ph / 50Hz	1800 × 1300 × 2600	1,120 kg
WHD - 100	1,000 CFM	1,700	7–16 bar g	4" NB Flange	220V / 1Ph / 50Hz	2000 × 1400 × 2700	1,450 kg
WHD - 150	1,500 CFM	2,550	7–16 bar g	5" NB Flange	220V / 1Ph / 50Hz	2300 × 1600 × 2900	2,100 kg
WHD - 200	2,000 CFM	3,400	7–16 bar g	6" NB Flange	220V / 1Ph / 50Hz	2600 × 1800 × 3100	2,850 kg

ENGINEERING SELECTION & OPERATIONAL NOTES

- Mandatory Pre-Filtration: Grade Y coalescing filter (0.01µm, residual oil <0.01 mg/m³) must be installed upstream to prevent desiccant oil poisoning.
- Mandatory After-Filtration: Grade P particulate filter (1µm) installed downstream to capture trace desiccant fines.
- Custom higher-pressure models up to 40 bar g available for PET bottle blow-molding installations.

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

- Pharmaceutical & Bulk API Manufacturing** Guarantees moisture-free sterile air for pill coating, fluidized bed drying, powder conveying, and cleanroom air handling.
- Automotive Robotics & High-Gloss Finish** Delivers bone-dry propellant air for automated electrostatic bells, preventing water-spotting and adhesion failure in OEM paint shops.
- Sub-Zero Outdoor Air Distribution** Eliminates catastrophic pipe freeze-up in exposed industrial pipelines during winter and cold-storage operations.
- Electronics & Semiconductor Fabrication** Prevents wafer oxidation, micro-corrosion, and static discharge defects during robotic chip placement and lead-frame assembly.
- Nitrogen / Oxygen Gas Generation** Protects carbon molecular sieve (CMS) beds in PSA gas plants from catastrophic water fouling and capacity decay.

Engineering Comparison: Win WHD vs. Conventional Heatless Dryers			Standard ISO 9001:2015 Quality
Engineering Parameter	Win Equipments Engineered Solution	Standard / Imported Benchmark	
Pressure Vessels	SA 516 Gr. 70 boiler plate, ASME / IS 2825 Class II certified	Lightweight commercial pipe with minimal corrosion allowance	
Switching Valves	Stainless steel pneumatic angle-seat valves (>1M cycle life)	Plastic diaphragm solenoid valves prone to tearing	
Purge Control	Optional Dew Point Dependent Switching (DDS) cuts purge <5%	Fixed 15% timer purge wasting continuous compressor power	
Acoustic Silencer	Heavy dual acoustic expansion mufflers (<75 dBA)	Small open felt mufflers (>85 dBA sound blast)	

MANUFACTURING STANDARDS & FACTORY VERIFICATION

Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

✓ Factory Acceptance Testing (FAT)

- 100% Radiographic / Ultrasonic non-destructive weld testing on pressure vessels
- 1.5× Hydrostatic pressure hold test witnessed by third-party inspection agencies
- Continuous 24-hour hygrometer dew point verification test prior to packaging
- PLC automated timing sequence and fail-safe valve closure validation

★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



Manufacturer Warranty Assurance: 36 Months structural warranty on pressure vessels; 12 months comprehensive on valves and electronics. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

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